

Name : Salma Mohamed Fouad

Supervised BY: dr. Ayman Alhelbawy

- **Part 2**

- 1. Preprocessing :**

Replace any digit with <num>

Removed any non-word char

- 2. Build N gram Models:**

Using “def build_word_model_with_padding()”:from lab

That create each model with every word with the (n-1)-words frequencies

- 3. Build models per category:**

For each category use “def build_word_model_with_padding()”

Saves:

ngram_counts: A frequency dictionary containing all observed n-grams and their occurrence counts within the category.

total_ngrams: The **total** number of n-grams extracted from the category corpus, used for probability normalization.

vocab: The vocabulary set containing all **unique** tokens observed in the training data for the category.

vocab_size: The size of the vocabulary ,used for smoothing techniques such as Laplace smoothing.

n: The order of the language model (e.g., unigram, bigram, trigram).

4. Probability | Perplexity :

- **Two classification criteria are supported:**

1. **Maximum log-likelihood**, which selects the class that assigns the highest probability to the input sequence.
2. **Minimum perplexity**, which selects the class for which the sequence is least surprising.

Perplexity provides a normalized measure of model fit, while log-likelihood reflects the raw sequence probability.

The vocabulary size $|V|$ is added in the denominator as part of Laplace smoothing. This adjustment is used to:

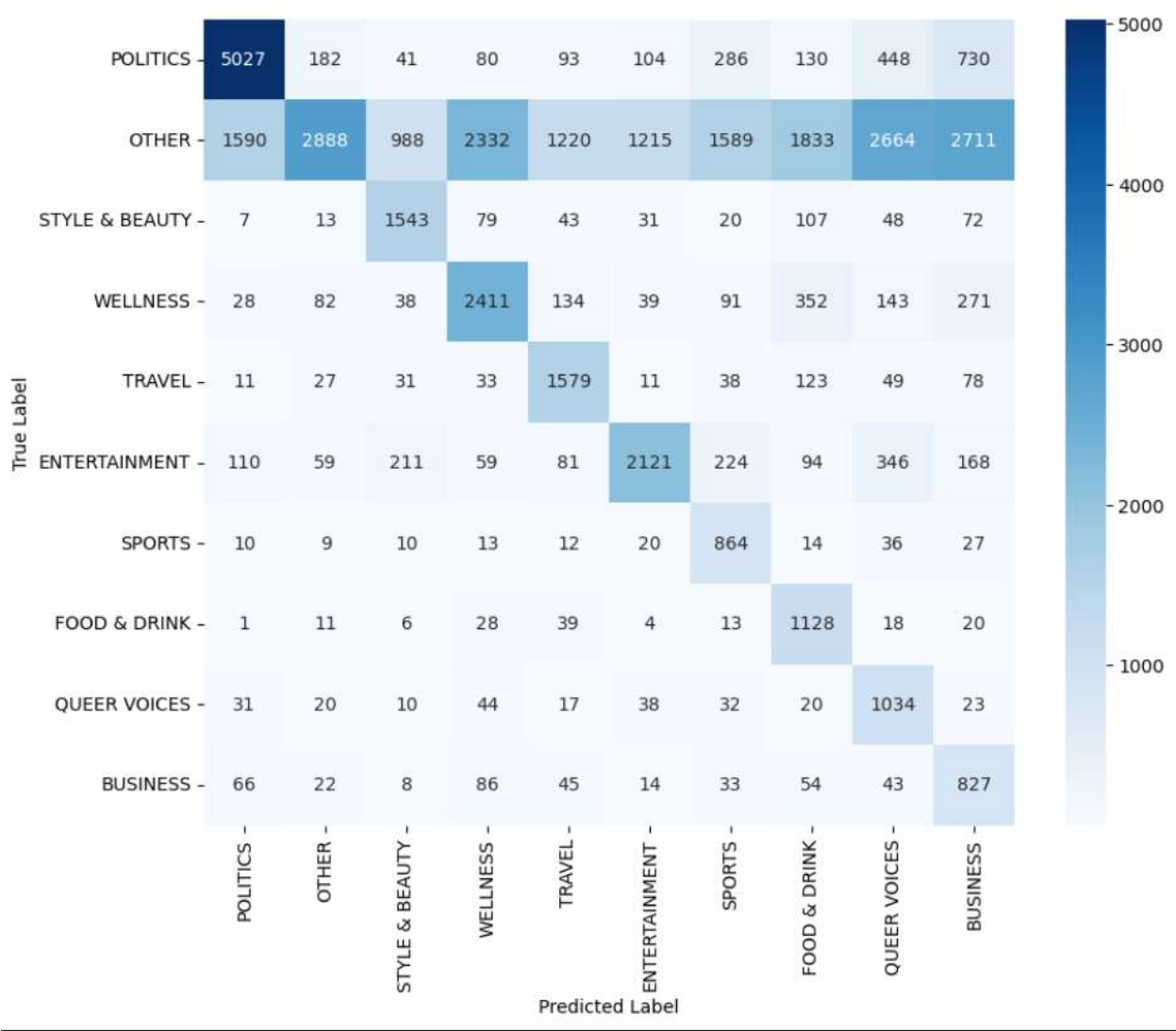
- **Avoid Zero Probabilities:** It prevents the model from assigning zero probability to unseen n-grams, which would otherwise make them impossible during inference.
 - **Improve Generalization:** By allocating probability mass to unseen or rare events, the model can better handle out-of-vocabulary words in new data.
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Tried 2 data (original vs OTHERresampled)

	Original	OTHERresampled
Unigram	0.4894	0.4635
Bigram	0.3164	0.3498
Trigram	0.1649	0.1458

Confusion matrices :

Original UNlgram:



OTHERresampled UNlgram:

